

Automatic Recognition of Quranic Tajweed Rules Across Riwayat: A Corpus and Machine-Learning Study of Idgham and Qalqalah **[TODO: confirm final title]**

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Abstract. Automatic assessment of Quranic recitation requires systems that determine not only *what* was recited but *how* it was pronounced. We present a riwayat-aware framework for the automatic recognition of Tajweed rules from speech. A corpus of 7,998 verse-level recordings (11.17 h) was built from MP3Quran recitations of Juz' 'Amma by fifteen reciters covering the riwayat of Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'; verse segments were obtained by forced alignment with manual reconciliation, and occurrences of Idgham and Qalqalah were localised through rule-based candidate extraction, manual acoustic annotation and expert verification (705 unique textual occurrences). Recitation segments are described by eight interpretable acoustic descriptors and classified into five Tajweed categories with SVM, random forest, k -nearest neighbours, XGBoost and a multilayer perceptron. The proposed representation outperforms an MFCC baseline with deltas for every classifier, reaching a Macro-F1 of 0.6935 with XGBoost and gains of up to +0.1348. Ablation identifies duration and a depth metric as the dominant cues, SMOTE raises Idgham bila-Ghunnah recall from 0.0678 to 0.4915, and cross-riwayat transfer yields Macro-F1 values of 0.6584–0.6718.

Keywords: Quranic recitation , Tajweed recognition , Idgham , Qalqalah , acoustic feature extraction , machine learning , riwayat-aware classification

1 Introduction

Qur'anic recitation is a highly structured form of oral Arabic whose pronunciation is governed by the principles of Tajweed and normally acquired through repeated practice with expert feedback. Conventional assessment therefore depends on expert listening, making frequent, scalable and objective evaluation difficult outside formal instruction. Recitation is further transmitted through recognised *riwayat*—narrations sharing the same Qur'anic text but differing in phonetic, articulatory and recitational detail—so a computational system must treat legitimate riwayat variation as distinct from pronunciation error. This study considers the “Big Three” riwayat: Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'.

Automatic assessment of Qur'anic recitation differs fundamentally from conventional speech recognition, which determines what was recited rather than how particular phonetic sequences were realised; reciter identification models speaker

characteristics, and generic pronunciation assessment provides no explicit rule-level representations. The central challenge is thus to detect specific Tajweed phenomena from the acoustic signal despite speaker variability, riwayat-related differences and scarce annotated resources—a challenge for which appropriately annotated speech data is a prerequisite that existing Qur'anic corpora do not meet (Sect. 2).

Idgham and Qalqalah offer complementary test cases. Idgham is an assimilation between adjacent consonants that modifies the temporal and spectral structure of the affected boundary, optionally with nasalisation (ghunnah); Qalqalah is an echo-like release of the consonants *qaf*, *ta'*, *ba'*, *jim*, *dal* in sukun states. Their contrasting signatures make them suitable for testing whether conventional acoustic descriptors capture rule-specific information while remaining robust across riwayat.

Prior work leaves four aspects insufficiently explored: the dominance of lexical- and speaker-level objectives over explicit Tajweed modelling; the scarcity of studies treating a rule occurrence—rather than an utterance—as the unit of analysis; the absence of rule-level temporal annotations; and the unexamined influence of multiple riwayat on acoustic realisation and detection. This paper therefore treats corpus construction, riwayat identification, rule-level annotation, feature extraction, supervised classification and evaluation as interconnected components of one problem.

The contributions are fourfold. *Resource*: a Qur'anic speech corpus covering the Big Three riwayat, annotated at the Tajweed-rule level. *Method*: a riwayat-aware pipeline connecting annotation with acoustic feature extraction and supervised classification. *Experimental*: a quantitative evaluation of five classifiers on Idgham and Qalqalah across the three riwayat. *Application*: a foundation for computer-assisted recitation systems providing rule-specific, interpretable feedback while respecting legitimate riwayat variation.

2 Related Work

2.1 Qur'anic Speech Resources and Recognition

Existing Qur'anic speech resources differ considerably in scale, speaker diversity, recording conditions, annotation granularity and intended use (Table 1). Verse-level repositories such as EveryAyah [1] link recordings to verses but provide no temporal Tajweed annotation, and the large-scale ASR corpora—the 581,216-recording Quran Recitations corpus [2] and the speaker-disjoint Quran-Ayah-Corpus [3]—prioritise transcription over pronunciation correctness. Diversity-oriented resources (Tarteel [4], Ar-DAD [5], QDAT [6] and the crowdsourced corpus of non-Arabic-speaking reciters [7]) contribute speakers, realistic conditions or expert judgements, but only text- or category-level supervision. Transmission-aware resources are the most recent step: AQQD [8] adds qira'at and phonetic-variation metadata and Riwaya-ID [9] separates Warsh from Hafs with 82% accuracy using wav2vec 2.0 [10] and Whisper [11], yet neither provides temporally localised rule labels. Collectively these resources trade lexical and speaker coverage against rule-level supervision, so

corpus size alone cannot satisfy the requirements of automatic Tajweed recognition.

Table 1. Comparison of Qur'anic speech resources. “Rule-level temporal labels” indicates whether the resource localises individual Tajweed phenomena in time.

Resource	Recordings	Speakers	Supervision	Rule-level
EveryAyah [1]	verse-level	professional	verse–audio link	no
Tarteel [4]	$\approx 25,000$ (67.39 h)	$> 1,200$	audio–text	no
Ar-DAD [5]	15,810 + 397	30 + 12	reciter identity	no
QDAT [6]	$> 1,500$	> 150	3 rules, expert	categorical
Crowdsourced [7]	$\approx 7,000$ (1,166 ann.)	1,287	6 categories	categorical
Quran Recitations [2]	581,216 ($\approx$ 2,882 h)	44	transcript	no
Quran-Ayah-Corpus [3]	263,263	44	transcript, disjoint splits	no
AQQD [8]	24,183	309	qira'at metadata	partial
Riwaya-ID [9]	> 700 h	–	riwayah label (82%)	no
Ours	7,998 (11.17 h)	15	rule-level, expert-verified	yes

2.2 Automatic Tajweed Recognition

Acoustic Tajweed assessment began with phoneme-duration analysis of Madd and Ghunnah [12] and progressed through MFCC-based statistical and neural classifiers for Qalqalah [13, 14], binary SVM formulations [15], convolutional [16] and recurrent [17] models, and hybrid ASR–classification pipelines [18] (Table 2). Three conclusions recur across these settings: Tajweed phenomena are not equally separable acoustically; accuracy alone is insufficient under class imbalance; and performance obtained under controlled conditions does not imply robust generalisation to unseen speakers, recordings or riwayat. Idgham and Qalqalah nonetheless remain fragmented across datasets, feature representations and evaluation protocols, with no unified framework spanning multiple reciters and transmission traditions.

Table 2. Prior automatic Tajweed recognition systems compared with the present study. Acc.\ = accuracy; val.\ = validation.

Study	Rules	Representation	Model	Data	Reported result
Mohammed et al. [12]	Madd, Ghunnah	duration	threshold	10 reciters	interpretable cues
Qalqalah VQ [13]	Qalqalah	MFCC	vector quant.	–	feasibility
Qalqalah MLP [14]	Qalqalah Kubra	MFCC	MLP	–	feasibility
Smartajweed [15]	4 binary rules	filter bank	RBF-SVM	–	99% val. acc.
Omran et al. [16]	Qalqalah	MFCC	CNN	3,322 / 4 rec.	90.8% val. acc.
Al-ahjal et al. [17]	3 rules	MFCC	LSTM	$>1,500$ (QDAT)	77.0/59.4/66.1%
TajweedAI [18]	Qalqalah	ASR + alignment	hybrid	internal/ext.	100% / 57.14%
Ours	2 Idgham + 3 Qalqalah	8 descriptors	5 classifiers	7,998 / 15 rec.	Macro-F1 0.6935

2.3 Research Gap and Positioning

Four gaps define the present work. First, large ASR corpora provide verse-level textual supervision whose objective—transcription—does not encode whether a rule was correctly realised. Second, rule-level temporal annotation identifying *where* a phenomenon occurs in continuous recitation is largely absent, preventing models from learning rule realisations independently of context. Third, rule-specific datasets trade speaker diversity against acoustic control: four professional reciters [16] versus crowdsourced but heterogeneous recordings [6]. Fourth, riwayat is treated as a global classification target [9] rather than as a factor shifting the acoustic realisation of individual rules. The proposed study addresses these gaps by formulating Idgham and Qalqalah recognition as a riwayat-aware, fine-grained acoustic learning problem on a rule-annotated corpus that retains reciter and riwayat identity, bridging large-scale Qur'anic ASR research and rule-specific Tajweed assessment.

3 Methodology

The framework treats automatic Tajweed recognition as a sequence of dependent stages (Fig. 1). Full-surah recordings are aligned and segmented into verse-level clips with explicit reciter and riwayat metadata (Sect. 4); Tajweed occurrences are localised acoustically through a hybrid rule-based, expert-verified annotation protocol (Sect. 5); segments are represented by interpretable acoustic descriptors and analysed for redundancy and imbalance (Sect. 6); and the resulting instances support

a five-class supervised task evaluated with complementary classifiers, ablations, balancing strategies, an MFCC baseline, cross-riwayah transfer and bootstrap uncertainty estimation (Sects. 7–9). Three principles govern the pipeline: the rule occurrence, not the verse, is the unit of analysis; riwayat identity is preserved at every stage so that legitimate variation can be separated from error; and every derived artefact remains linked to the original signal and to the canonical text.

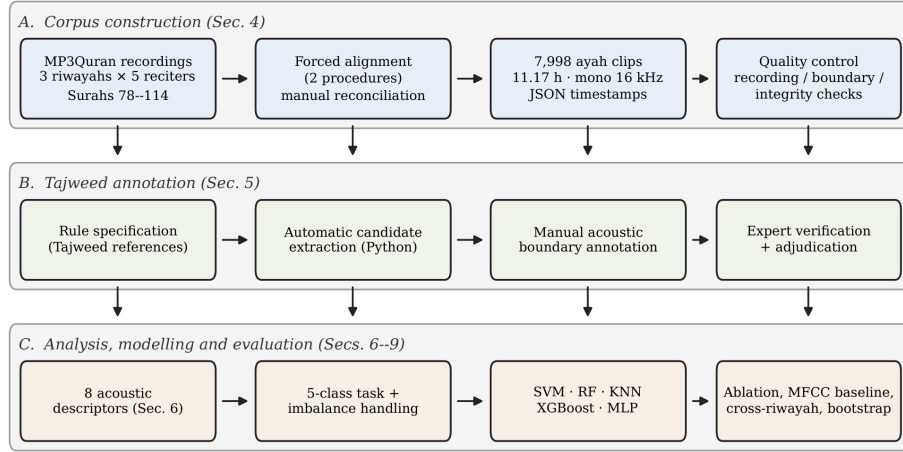


Fig. 1. Overview of the riwayat-aware Tajweed recognition framework: corpus construction with forced alignment and quality control (Sect. 4), hybrid Tajweed annotation (Sect. 5), and acoustic analysis, modelling and evaluation (Sects. 6–9).

4 Corpus Construction

4.1 Source Material and Riwayah Coverage

Audio was collected from the publicly available MP3Quran repository of complete Qur'anic recitations [19]. Rather than restricting the corpus to one tradition, three riwayat were included—Hafs 'an 'Asim, Warsh 'an Nafi', and Qalun 'an Nafi'—each represented by five reciters, giving fifteen reciters in total. Reciter selection prioritised inter-speaker variability (vocal characteristics, articulation, rate, prosody) within a well-defined tradition, because a model trained on a single reciter may learn speaker-specific rather than rule-specific characteristics. The scope is Juz' 'Amma (surahs 78–114), i.e. 37 surahs and 564 ayat per reciter, targeting 555 full-surah recordings; the canonical content corresponds to 8,460 reciter-ayah instances, of which approximately 14.14 h of full-surah audio were collected.

4.2 Segmentation, Alignment and Quality Control

Complete surah recordings were processed with two automatic forced-alignment procedures whose timestamps and recognised text were retained separately and reconciled manually into a consolidated segmentation file per surah, from which ayah-level clips were extracted; this prevents alignment or recognition errors from propagating unchecked into the training data. Quality control operated at three levels: recording level (availability, readability, correspondence with the intended surah and reciter); segmentation level (cross-method comparison and review of boundaries); and data-integrity level (expected surah and ayah counts checked against the canonical structure, with missing recordings identified explicitly rather than treated as complete). Alignment confidence scores are stored to flag potentially unreliable segments.

4.3 Corpus Organisation and Statistics

The corpus separates original recordings from derived segments and metadata: per-riwayah directories contain, for each reciter, an `audio/` folder of full-surah recordings, an `ayat/` folder of verse segments, and JSON timestamp files linking every segment to its source position. After segmentation, alignment and validation, 7,998 verse-level recordings were retained—94.5% of the theoretical maximum—comprising 11.17 h of aligned audio distributed evenly across riwayat (Table 3); for modelling, audio is decoded to mono at 16 kHz. The longer mean verse duration of Warsh and Qalun relative to Hafs is consistent with differences in recitation tempo and in the realisation of prolongation (madd) rules.

Table 3. Composition of the verse-level corpus by riwayat.

Riwayah	Reciters	Recordings	Duration (h)	Mean verse dur.\ (s)
Hafs 'an 'Asim	5	2,739	3.57	4.69
Warsh 'an Nafi'	5	2,562	3.78	5.30
Qalun 'an Nafi'	5	2,697	3.83	5.11
Total	15	7,998	11.17	5.03

Rule extraction from the canonical text identified 328 Idgham and 377 Qalqalah occurrences within Juz' 'Amma—705 unique textual occurrences before replication across reciters and riwayat (Table 4). Because each occurrence is realised by up to fifteen reciters, the number of acoustic instances available for learning is substantially larger than the textual counts, enabling the same phenomenon to be studied under diverse speakers and riwayat. Recording duration is balanced across riwayat (32–34% each), whereas subtype frequencies are strongly skewed: Lam Shamsiyyah accounts for more than 72% of Idgham occurrences while Idgham Lam-Ra occurs once, and Qalqalah Kubra (12 occurrences) is far rarer than Qalqalah Wusta (145).

Table 4. Unique textual occurrences of the annotated Tajweed subtypes in Juz' 'Ammah. Subtypes marked † constitute the five-class recognition task of Sect. 7; the remainder are annotated but excluded from the present classification experiments[**TODO: confirm subtype-exclusion criteria**].

Idgham subtype	Occ.	Qalqalah subtype	Occ.
Lam Shamsiyyah	237	Qalqalah Wusta [†]	145
Idgham bi-Ghunnah [†]	65	Qalqalah Sughra [†]	123
Idgham bila-Ghunnah [†]	25	Qalqalah Mushaddadah [†]	97
Idgham Lam-Ra	1	Qalqalah Kubra	12
Total	328	Total	377

5 Annotation Methodology

Annotation provides rule-specific labels at the level of individual ayat. The protocol derives from established Tajweed references, whose linguistic and phonetic descriptions were translated into explicit, consistently applicable annotation criteria, and combines automatic candidate identification with manual acoustic verification and expert review.

5.1 Idgham

Idgham denotes the assimilation of one consonant into a following consonant, here primarily nun sakinah and tanwin before the letters designated for Idgham, optionally with nasalisation. Acoustically it is expected to remove or reduce an independent realisation of the first consonant, increase temporal overlap between adjacent segments, and yield a boundary dominated by the following consonant, with added low-frequency energy and nasal resonance when ghunnah is present. Candidates were extracted from the Qur'anic text by a Python script encoding the formal Tajweed conditions; the waveform and spectral representation of each corresponding recording were then inspected to localise the realisation and confirm the assimilation. Boundaries were placed to capture the acoustic event while excluding unrelated segments, with particular care where coarticulation, recording quality, rate or context made the boundary ambiguous.

5.2 Qalqalah

Qalqalah is the echo-like release produced by *qaf*, *ta'*, *ba'*, *jim*, *dal* in states requiring the rule, notably sukun states, without an independent vowel. Its signature is a distinct consonantal release: a transient energy increase, a clear burst, rapid spectral change and a short transition into the neighbouring segment. Candidates were identified textually from the relevant letters and contextual conditions, after which the temporal region of the release was located manually and verified as eligible.

Because the realisation varies with reciter, articulation, recording conditions and context, acoustic evidence was always interpreted jointly with the textual condition, and genuine Qalqalah was distinguished deliberately from ordinary consonantal release.

5.3 Workflow, Consistency and Validation

The workflow comprises four steps: *rule specification*, converting Tajweed definitions into annotation criteria; *automatic candidate extraction* from the textual material; *manual acoustic annotation* of temporal boundaries and acoustic parameters; and *expert verification* of rule applicability and segmentation. Consistency was enforced by deterministic textual rules for candidate generation and by predefined boundary criteria applied uniformly. All labels then underwent expert review, with uncertain boundaries, weak realisations and ambiguous conditions examined individually; an annotation was retained only when linguistic condition and acoustic realisation were judged compatible. Inter-annotator agreement was assessed over categorical identification and temporal localisation, with disagreements resolved by expert adjudication[**TODO: report quantitative inter-annotator agreement statistic**]. The ground truth thus combines rule-based extraction, acoustic localisation and expert judgement—essential because a rule's presence in the text does not determine its acoustic realisation.

6 Data Analysis

6.1 Acoustic Feature Representation

Each annotated segment is described by eight handcrafted descriptors chosen to reflect the phonetic mechanisms of the target rules: RMS mean (signal energy), zero-crossing-rate (ZCR) mean (temporal fine structure), spectral-centroid mean (spectral distribution), spectral-flux mean (spectral change), mean and peak nasality (nasal coupling), a depth metric (relative prominence and internal structure of the acoustic event), and duration (temporal extent of the rule-related event). Segment-level statistics are used so that every descriptor is defined uniformly across instances of differing length, and each corresponds to an acoustic dimension that Tajweed theory predicts to be affected by assimilation, nasalisation or consonantal release (Table 5).

Table 5. The eight acoustic descriptors and their relation to Tajweed realisation.

Descriptor	Acoustic dimension	Tajweed relevance
RMS mean	signal energy	overall intensity of the rule event
ZCR mean	temporal fine structure	noise-like transition at assimilation
Spectral-centroid mean	spectral distribution	brightness change at release
Spectral-flux mean	spectral change	rapid transition of Qalqalah release
Mean nasality	average nasal coupling	ghunnah of Idgham bi-Ghunnah
Peak nasality	maximal nasal coupling	strength of nasalisation
Depth metric	prominence / internal structure	articulatory depth of the event
Duration	temporal extent	assimilation and prolongation timing

6.2 Redundancy, Normalisation and Cleaning

Correlation analysis indicates partial redundancy: the correlation between ZCR and spectral centroid reaches 0.98, mean and peak nasality correlate at approximately 0.63, and spectral centroid and spectral flux show a moderate positive relationship. Correlated descriptors are not interchangeable, but redundancy motivates normalisation and subset evaluation. Features are standardised as

$$Z_j = \frac{x_j - \mu_j}{\sigma_j}, \quad (1)$$

with μ_j and σ_j estimated from the training partition only, which is essential for distance- and margin-based models[**TODO: confirm the exact normalisation implementation and frame-level extraction parameters**]. A cleaning stage then removed unreliable observations, reducing the instance count in all five classes, most visibly in the majority classes, while Idgham bila-Ghunnah remained substantially under-represented[**TODO: report exact pre-/post-cleaning class counts and cleaning criteria**].

Class distribution. At corpus level the riwayat balance limits riwayat-specific bias, but the subtype distribution is markedly imbalanced: models optimised for overall accuracy may favour dominant classes while failing on rare but linguistically important phenomena, so evaluation must rely on class-sensitive metrics alongside accuracy (Sect. 7.4). The multi-reciter, multi-riwayat design additionally permits testing whether learned representations capture Tajweed-related rather than reciter- or riwayat-specific patterns.

7 Machine Learning Framework

7.1 Task Formulation

Recognition is formulated as supervised multiclass classification: each segmented instance is assigned to one of five Tajweed classes—Idgham bi-Ghunnah, Idgham bila-Ghunnah, Qalqalah Mushaddadah, Qalqalah Sughra and Qalqalah Wusta—from the descriptors of Sect. 6. These five classes are the subtypes of Table 4 retained for the classification experiments; the remaining annotated subtypes lie outside the present experimental scope.

7.2 Classifiers

Five complementary algorithms cover distinct learning paradigms. The soft-margin support vector machine (SVM) minimises $\frac{1}{2}\|\mathbf{w}\|^2 + c\sum \xi_i$ subject to $y_i(\mathbf{w}^T\phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$, combining binary decision functions for the five-class task; it suits compact descriptors but is sensitive to scaling and kernel choice. The random forest (RF) aggregates bootstrap-trained trees over randomised feature subsets,

$$\hat{y} = \text{mode}\{T_b(\mathbf{x})\}_{b=1}^B, \quad (2)$$

modelling nonlinear descriptor interactions and providing importance estimates. k -nearest neighbours (KNN) assigns the majority class among the k closest training observations,

$$\hat{y} = \text{mode}\{y_i : \mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})\}, \quad (3)$$

serving as an instance-based baseline sensitive to local acoustic similarity. XGBoost [20] builds an additive ensemble $\hat{y}_i = \sum f_t(\mathbf{x}_i)$ over regression trees, optimising a regularised objective $\mathcal{L} = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_t)$, which captures higher-order interactions among temporal, spectral and phonatory descriptors while remaining effective for small tabular feature sets. Finally, a multilayer perceptron (MLP) applies hidden-layer transformations $\mathbf{h} = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$ with a softmax output $P(y = k|\mathbf{x}) = \exp(z_k) / \sum \exp(z_i)$, testing whether nonlinear combinations of the descriptors add discriminative information. Together these models evaluate whether the acoustic representation remains effective across substantially different learning paradigms.

7.3 Class-Imbalance Handling

Imbalance was investigated through five configurations: unbalanced training, class weighting, SMOTE [21], Borderline-SMOTE, and SMOTE with class weighting. SMOTE synthesises minority observations by interpolating between neighbouring minority samples,

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda(\mathbf{x}_{\text{nn}} - \mathbf{x}_i), \lambda \in [0, 1], \quad (4)$$

whereas Borderline-SMOTE restricts synthesis to samples near class boundaries[**TODO: report SMOTE neighbour count and sampling ratio**]. All configurations are evaluated with conventional and imbalance-sensitive metrics, since high overall accuracy can coexist with poor minority-class recognition.

7.4 Evaluation Metrics

Performance is measured by accuracy, per-class precision, recall and F1-score, Macro-F1, weighted F1, balanced accuracy and one-vs-rest ROC-AUC. With $K = 5$ classes, Macro-F1 is the unweighted mean of per-class F1-scores and balanced accuracy the mean of per-class recalls; precision–recall (PR) analysis complements ROC analysis because ROC-AUC can remain high for rare classes even when precision deteriorates at practical operating points.

8 Experiments

8.1 Experimental Protocol

Seven experiment families address the research questions: **E1** evaluates the five classifiers on the cleaned dataset with the proposed representation; **E2** compares it against an MFCC + delta + delta-delta baseline under identical protocols; **E3** analyses per-class behaviour through normalised confusion matrices; **E4** examines discriminability with one-vs-rest ROC and PR curves; **E5** quantifies descriptor contributions through tree-based importance and ablation; **E6** compares the imbalance-handling configurations of Sect. 7.3; and **E7** tests cross-riwayat generalisation by training on one riwayat and evaluating on another for all six ordered pairs. Uncertainty is estimated by bootstrap resampling of the test set (1,000 resamples). Table 6 summarises the design.

Table 6. Experimental design: question, protocol and primary metrics of each experiment family.

ID	Question	Protocol	Primary metrics
E1	Does the representation support recognition?	5 classifiers, cleaned data	Acc., bal. acc., Macro-F1
E2	Is it better than MFCC?	identical protocol, MFCC baseline	Macro-F1, Acc.
E3	Which classes drive performance?	normalised confusion matrices	per-class recall
E4	How separable are the classes?	one-vs-rest ROC / PR	ROC-AUC, PR
E5	Which descriptors matter?	importance + ablation	Macro-F1 change
E6	Does balancing help?	5 training configurations	bal. acc., minority recall
E7	Does it transfer across riwayat?	6 train→test pairs	Macro-F1

Protocol details to be finalised. A held-out test set provides final evaluation and 5-fold cross-validation is used for the unbalanced-versus-SMOTE comparison. The train/test proportion, the splitting unit (segment, verse, surah or reciter), stratification and the random seed must be documented to exclude speaker- or recording-level leakage, a particular risk for recitation data[**TODO: specify split strategy, proportion, stratification and seed**]. Hyperparameter grids, selection criteria and final configurations[**TODO: report hyperparameter search and final configurations**], with software versions and hardware[**TODO: report Python/library versions and compute environment**], are likewise required for reproducibility.

9 Results

9.1 Overall Classifier Performance

Table 7 reports the primary comparison (class-weighted configuration). XGBoost achieves the highest Macro-F1 (0.6935) with accuracy 0.7202 and balanced accuracy 0.6953; SVM attains the highest balanced accuracy (0.7126) and random forest the highest accuracy (0.7298), while MLP is weakest on Macro-F1 (0.6219). The divergence between accuracy and Macro-F1—random forest leads on accuracy yet trails XGBoost by 0.0252 on Macro-F1—confirms that accuracy alone mischaracterises this imbalanced problem.

Table 7. Classifier comparison on the held-out test set with the proposed representation (class-weighted configuration). Macro-recall equals balanced accuracy by definition and is reported once.

Model	Accuracy	Balanced acc.	Macro-prec.	Macro-F1	Weighted F1
SVM	0.7083	0.7126	0.6902	0.6920	0.7080
Random forest	0.7298	0.6668	0.7036	0.6683	0.7192
KNN	0.7179	0.6632	0.6872	0.6687	0.7125
XGBoost	0.7202	0.6953	0.6951	0.6935	0.7195
MLP	0.7071	0.6356	0.7001	0.6219	0.6873

9.2 Proposed Representation Versus MFCC Baseline

The proposed descriptors improve Macro-F1 for every classifier (Fig. 2), with the largest gains for random forest (+0.1348), KNN (+0.1336) and XGBoost (+0.1191); accuracy rises from 0.5869 to 0.7298 (random forest), 0.5774 to 0.7179 (KNN) and 0.6119 to 0.7202 (XGBoost). Consistency across all five paradigms indicates that the advantage originates in the information content of the representation rather than in an interaction with a particular decision boundary.

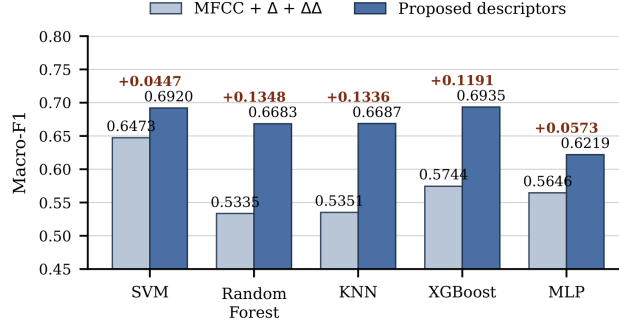


Fig. 2. Macro-F1 of the proposed acoustic descriptors and of the MFCC + delta + delta-delta baseline for each classifier; annotations give the absolute gain of the proposed representation.

9.3 Per-Class Behaviour, Confusion and Discriminability

Per-class results differ sharply (Fig. 3). Idgham bi-Ghunnah is recognised almost perfectly by every classifier (recall ≈ 1.00 , ROC-AUC 1.00), and Qalqalah Wusta remains strong and architecturally stable across all five models (recall 0.88–0.92, ROC-AUC 0.99). Qalqalah Mushaddadah is intermediate (recall 0.58–0.72, ROC-AUC 0.89) and Qalqalah Sughra is recovered reliably by the strongest model (recall 0.55, ROC-AUC 0.84). Idgham bila-Ghunnah is the most difficult class, with recall spanning 0.017–0.49 across models and ROC-AUC 0.87 for XGBoost; its errors are distributed across several competitors rather than concentrated in one, indicating overlap with multiple categories under the current representation. SVM's relatively high recall for Idgham bila-Ghunnah (0.49) explains its leading balanced accuracy despite XGBoost's higher Macro-F1. PR analysis shows precision for Idgham bila-Ghunnah falling steeply as recall increases, so its ROC-AUC must not be read as uniform strength: the class is ranked above chance, yet only part of it can be recovered without admitting many false positives.

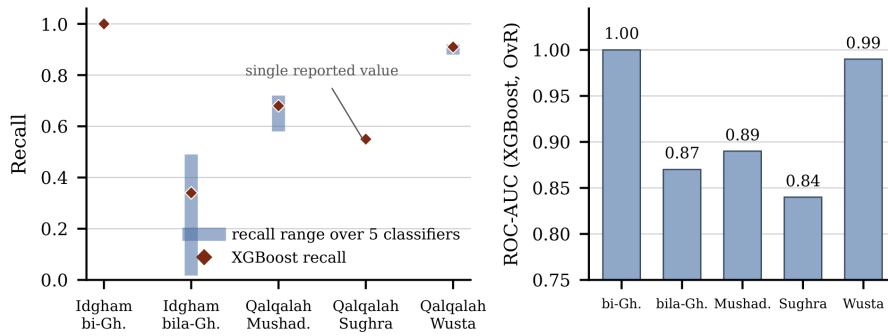


Fig. 3. Per-class discriminability of the five Tajweed categories, with the Idgham subtypes followed by the Qalqalah subtypes. Left: recall attained by the five classifiers, shown as the range across models with the XGBoost value marked; Qalqalah Sughra carries a single reported value. Right: ROC-AUC of XGBoost for the same classes.

9.4 Feature Importance and Ablation

In XGBoost, duration dominates feature importance (≈ 0.52) followed by the depth metric (≈ 0.21); in random forest the same two descriptors lead (depth ≈ 0.33 , duration ≈ 0.24) over a broader distribution. Ablation confirms their causal contribution (Fig. 4): removing the depth metric reduces Macro-F1 from 0.6935 to 0.5169 (-0.1766) and removing duration to 0.5587 (-0.1348), whereas retaining duration and depth alone still yields 0.6424 and nasality alone collapses to 0.2290, i.e. no better than random over five classes. Removing the spectral descriptors (centroid and flux) costs only 0.0423 Macro-F1, and removing the strongly correlated nasality pair 0.0391, so both groups carry information that complements—but does not duplicate—the temporal descriptors. The full set remains superior to every subset and every removal variant: the descriptors are not interchangeable, a small subset carries most of the discriminative signal, and the residual classes contribute the difference between a good and the best result.

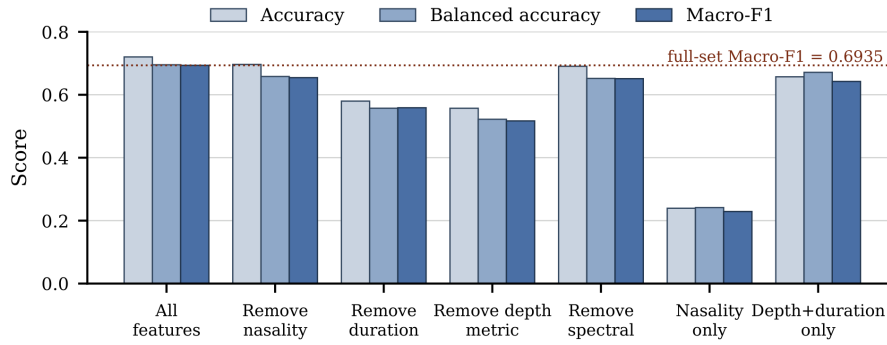


Fig. 4. Feature ablation with XGBoost on the held-out test set. Macro-F1, balanced accuracy and accuracy for seven configurations: the full descriptor set, four removal variants and two restricted subsets. Protocol, seed and class weighting are identical throughout.

Class-balancing results. Balancing trades overall accuracy for minority-class sensitivity (Fig. 5). Unbalanced training yields the highest accuracy (0.7357) but an Idgham bila-Ghunnah recall of only 0.0678 (F1 0.1111). Class weighting raises that recall to 0.3390 and Macro-F1 to 0.6935 at a cost of 1.55 points of accuracy, and SMOTE gives the strongest balanced performance: balanced accuracy 0.6617 $\rightarrow$ 0.7121, Macro-F1 0.6592 $\rightarrow$ 0.6951, and Idgham bila-Ghunnah recall 0.0678 $\rightarrow$ 0.4915 (+42.4 points). Borderline-SMOTE is less effective than standard SMOTE, and adding class weighting after SMOTE yields no further change, indicating that boundary-focused synthesis does not suit this feature space.

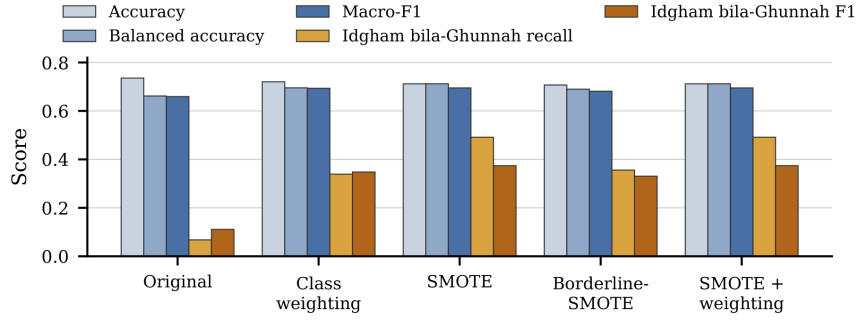


Fig. 5. Effect of class-balancing strategies with XGBoost. Accuracy, balanced accuracy, Macro-F1 and the recall and F1 of the rarest class, Idgham bila-Ghunnah, for the original training set, class weighting, SMOTE, borderline-SMOTE and SMOTE combined with class weighting.

9.5 Cross-Riwayah Generalisation and Uncertainty

Training on one riwayat and testing on another yields Macro-F1 between 0.6584 and 0.6718 across the six transfer directions (Fig. 6), a narrow range of 0.0134 with the best transfer Qalun $\rightarrow$ Warsh (0.6718) and the weakest Qalun $\rightarrow$ Hafs (0.6584): reasonably stable, yet consistently below the within-distribution XGBoost result, indicating partially riwayat-independent information alongside residual distribution shift. Bootstrap resampling gives a mean Macro-F1 of 0.6934 with a 95% confidence interval of [0.6604, 0.7240]. McNemar (XGBoost vs. SVM) and Wilcoxon (original vs. SMOTE, 5-fold) tests are specified in the analysis plan but their statistics and p -values are unavailable, so no claim of statistical superiority is made[**TODO: insert McNemar and Wilcoxon statistics and p-values**].



Fig. 6. Cross-riwayah generalisation (Macro-F1): rows are training riwayat and columns testing riwayat. Diagonal cells (within-riwayah evaluation) correspond to the pooled results of Table 7 and are not repeated here.

10 Discussion

10.1 Representation Quality Over Model Complexity

The results indicate that representation quality, rather than classifier complexity, primarily determines performance. MFCCs approximate the short-term spectral envelope and are strong general-purpose descriptors, but Tajweed realisation depends on temporal organisation, articulatory configuration, nasal coupling and consonantal release—dimensions that cepstral coefficients encode only indirectly. That the proposed descriptors improved every classifier, from instance-based KNN to boosted trees and a neural network, supports the interpretation that the gain resides in the acoustic evidence itself. The finding does not establish MFCCs as universally inferior; it shows that, for this task, a compact linguistically motivated descriptor set carries more relevant information than the evaluated MFCC configuration, and that model sophistication cannot compensate for a representation omitting linguistically relevant dimensions.

10.2 Duration and Depth as Acoustic Evidence of Tajweed

Duration and the depth metric converge across two independent forms of evidence—tree-based importance and controlled removal—making an incidental explanation unlikely. This is phonetically plausible: assimilation alters the temporal distribution of energy across a boundary and Qalqalah alters the release behaviour of a consonant, so both produce measurable temporal and structural consequences. More broadly, their prominence supports treating Tajweed as an acoustically observable realisation of phonetic instructions rather than a purely symbolic property of the text—the premise on which automatic assessment depends. Because duration is also sensitive to annotation boundaries (Sect. 11), its importance should be interpreted jointly with the boundary-uncertainty analysis.

10.3 Why Idgham bila-Ghunnah Remains Difficult

Idgham bi-Ghunnah supplies an explicit nasal cue that nasality descriptors capture directly, whereas Idgham bila-Ghunnah lacks that marker and must rely on subtler temporal and spectral assimilation cues that overlap with neighbouring phonetic events and other target classes. The contrast illustrates the distinction between phonetic definition and acoustic separability: rules unambiguous in theory may partially overlap in measured realisation. Scarcity compounds the problem, and the improvement obtained through SMOTE shows that part of the difficulty is statistical rather than purely acoustic; residual confusion after balancing therefore points to genuine acoustic overlap that no classifier change within the current representation is likely to remove.

10.4 Generalisation Across Riwayat as Domain Shift

The stability of cross-riwayah Macro-F1 suggests that the descriptors capture rule-related information not tied to a single tradition—encouraging for practical

assessment, which must accept recitations in different riwayat. The consistent degradation relative to within-distribution performance, however, shows that riwayat-independent classification cannot be assumed: articulation, timing, melodic realisation and recording conditions shift the feature distributions of the same category, particularly for duration- and prosody-sensitive descriptors. Cross-riwayat robustness should therefore be a design objective, with riwayat-related variability treated as a concrete source of domain shift for adaptation or invariant-representation methods.

10.5 Implications for Research and Practice

Two implications follow. Methodologically, Tajweed recognition should be modelled as a structured, rule-aware problem: because categories differ in articulatory mechanism, temporal realisation and acoustic salience, a representation preserving both shared and rule-specific dimensions is preferable to a single undifferentiated vector, and domain knowledge remains valuable even with conventional learners. Practically, interpretable descriptors enable localised, confidence-aware feedback—a suspected rule occurrence, its temporal location and the relevant pronunciation dimension—supporting computer-assisted tutoring beyond global scoring. Such predictions must be presented as assistive feedback rather than definitive religious or pedagogical judgement, with expert supervision retained for difficult cases.

11 Limitations

Corpus scope. The corpus covers Juz' 'Amma and two rule families only; observed characteristics may not represent the wider Qur'an, and fifteen reciters across three riwayat cannot establish corpus-independent generalisation, so reported gains are evidence within the experimental domain. **Imbalance and synthetic oversampling.** Natural subtype frequencies are skewed, and SMOTE synthesises feature-space samples rather than authentic recitations; gains from oversampling partly reflect better feature-space coverage, not additional phonetic diversity, so class-aware acquisition of genuine minority examples remains necessary. **Annotation uncertainty.** Assimilation phenomena have gradual acoustic transitions, so boundaries are inherently less discrete; because duration and depth depend on boundary placement, small shifts can alter feature values for acoustically equivalent recitations. Quantitative multi-expert agreement analysis and probabilistic boundaries would strengthen reliability assessment. **Acoustic representation.** Segment-level statistics cannot represent long-range dependencies or transitions between neighbouring phonemes, precisely where assimilation evidence may reside; sequential or attention-based architectures should be evaluated alongside—not simply instead of—the interpretable descriptors. **External and statistical validity.** Results derive from one corpus and configuration, so without speaker-independent and external-corpus validation high in-domain performance does not establish

robustness to unseen speakers or conditions; and without McNemar and Wilcoxon statistics, superiority claims remain provisional.

Future directions. Three follow-ups address these limits directly. First, evaluating the identical protocol on the Hafs-only subset of an independent public recitation corpus would separate corpus-specific from representation-specific behaviour. Second, a riwayat-conditioned model—with the reciter's riwayat supplied as an explicit input or with per-riwayat fine-tuning—should reduce the cross-riwayat penalty without enlarging the balanced subset. Third, phoneme-aligned rather than fixed-window segmentation would remove the boundary noise that currently caps the recall of Qalqalah Mushaddadah and would make the duration descriptor directly interpretable in terms of Tajweed timing.

12 Conclusion

This paper formulated automatic Tajweed recognition as a riwayat-aware, fine-grained acoustic classification problem and instantiated it end to end: a 7,998-recording, 11.17 h corpus of Juz' 'Amma spanning Hafs, Warsh and Qalun; expert-verified rule-level annotation of Idgham and Qalqalah; an interpretable eight-descriptor representation; and a five-class evaluation across five learning paradigms. The representation outperformed an MFCC baseline for every classifier (best Macro-F1 0.6935, XGBoost), ablation established duration and the depth metric as dominant cues, balancing materially improved the difficult Idgham bila-Ghunnah class, and cross-riwayat transfer remained within 0.6584–0.6718. Future work will expand the corpus in surahs, reciters, riwayat and rules; quantify annotation agreement; adopt speaker- and corpus-independent evaluation; and investigate sequential, attention-based and multimodal models combining acoustic evidence with textual and phonological constraints, towards interpretable, confidence-aware recitation support that complements expert instruction.

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